# The Cycle Class of a Subvariety and the Reverse Arrow of the Hodge Conjecture

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This note typesets, without omission, the author's written account of the cycle class map and of what remains open in the Hodge conjecture.
It does not add a construction of the inverse.
The Clay statement is rational. The integral Hodge conjecture is stated separately below and is not claimed as a theorem of this note.

The last geometric object on screen is not a mystery class. It is the cycle class of a subvariety: the one arrow that is already a theorem. The video's closing line, "Decoding still required," refers to the reverse arrow, which remains open.

## What the final object is

In Step 05 the film names a closed algebraic subset

$$Z \subset X$$

of complex codimension $k$, and then writes its class

$$[Z] \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X).$$

That symbol $[Z]$ is the cycle class of $Z$. It is the source of the "balanced signal." Every later frame — the conjecture, the scoreboard, and the card "Hodge classes $\to$ algebraic cycles" — is asking whether an arbitrary Hodge class is a rational combination of objects of this same kind.

The yellow loop drawn on the wireframe is a picture, not a construction. A genuine algebraic subvariety of complex codimension $k$ is cut out by polynomial equations and has complex dimension $\dim X - k$. On an elliptic curve the algebraic classes in positive codimension are points, not real circles. The animation is showing the role of $Z$, not a particular equation.

## How the class is produced

Let $X$ be a smooth complex projective variety of dimension $n$, and let $Z \subset X$ be a closed irreducible subvariety of codimension $k$. The complex structure orients the regular locus of $Z$ as a real $2(n-k)$-dimensional manifold. That orientation supplies a fundamental class in homology,

$$[Z]_{\mathrm{hom}} \in H_{2n-2k}(X, \mathbb{Z}).$$

Poincaré duality on the compact oriented manifold $X$ converts it into a cohomology class

$$[Z] \;:=\; \mathrm{PD}([Z]_{\mathrm{hom}}) \in H^{2k}(X, \mathbb{Z}) \subset H^{2k}(X, \mathbb{Q}).$$

Equivalently, $Z$ defines a closed current of integration: for a smooth form $\varphi$ of degree $2n-2k$,

$$T_Z(\varphi) \;:=\; \int_{Z^{\mathrm{reg}}} \varphi.$$

The current $T_Z$ is closed, of type $(k,k)$, and represents the same class. Extending by linearity from irreducible subvarieties to rational algebraic cycles gives the cycle class map

$$\mathrm{cl}\colon \quad Z^k(X)_{\mathbb{Q}} \;\longrightarrow\; H^{2k}(X, \mathbb{Q}).$$

The image of cl is what the video calls the algebraic classes. The integral cycle-class map, with values in $H^{2k}(X, \mathbb{Z})$, is the same construction before tensoring with $\mathbb{Q}$.

## Why the class is automatically a Hodge class

A current of type $(k,k)$ pairs to zero against every form of type $(p,q)$ with $(p,q) \neq (n-k, n-k)$. Passing to cohomology, the class therefore has vanishing off-diagonal Hodge components:

$$[Z] \in H^{k,k}(X).$$

Combined with rationality,

$$[Z] \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X).$$

That intersection is the definition of a Hodge class. This is the easy arrow on screen:

$$\text{geometry } Z \;\longrightarrow\; \text{code } [Z].$$

It holds for every algebraic cycle, with no restriction on $k$. The film's line "Geometry $\rightarrow$ code. Always. No exceptions." is this theorem.

In codimension one the same class has a second, more explicit name. If $Z$ is a divisor, then

$$[Z] \;=\; c_1(\mathcal{O}_X(Z)),$$

the first Chern class of the associated line bundle. The exponential sequence produces every integral $(1,1)$ class in this way, and GAGA promotes the holomorphic bundle to an algebraic divisor. That is why $k = 1$ is settled over $\mathbb{Z}$, not only over $\mathbb{Q}$. There is no

analogous exponential sequence for $k \geq 2$.

## What "decode" would mean

The cycle class map always runs in one direction. Decoding the final class would mean inverting it on the Hodge subspace: given

$$\gamma \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X),$$

produce finitely many subvarieties $Z_i \subset X$ of codimension $k$ and rationals $a_i \in \mathbb{Q}$ such that

$$\gamma \;=\; \sum_i a_i [Z_i].$$

That is the Hodge conjecture, the Clay statement. The coefficients are rational. The last card of the video is exactly this request:

$$\text{Hodge class } \gamma \quad \longrightarrow \quad \text{algebraic cycle } Z.$$

The map cl is known. A section of cl over the Hodge classes is not known, except in the cases already listed in the film:

- $k = 1$, by Lefschetz $(1,1)$, integrally;
- $k = \dim X - 1$, by hard Lefschetz applied to the previous case;
- $\dim X \leq 3$, because those two degrees exhaust the interesting even cohomology.

In every remaining case — beginning with a codimension-2 class on a fourfold — the class $[Z]$ can be written down whenever $Z$ is given, and cannot in general be reconstructed from $\gamma$ alone. That is what "Decoding still required" refers to. It is not a missing formula for $[Z]$. It is the missing inverse over $\mathbb{Q}$.

## One-line decode

The final subvariety class in the video is

$$[Z] \;=\; \mathrm{PD}\big([Z]_{\mathrm{hom}}\big) \;\in\; H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X),$$

the Poincaré dual of the fundamental class of an algebraic cycle. It is the source of every balanced signal that geometry can produce. The open problem is not to interpret that class. The open problem is to decide whether every balanced rational signal arises in this way.

## The first open case

Let $X$ be a general fourfold and let $\gamma \in \mathrm{Hdg}^2(X)$ be arbitrary. The missing object is a finite collection of surfaces $Z_i \subset X$ and rationals $a_i \in \mathbb{Q}$ such that

$$\gamma \;=\; \sum_i a_i [Z_i].$$

That equality is the Hodge conjecture in its first open case. Writing the $Z_i$ down from $\gamma$ would be a section of cl. The cycle class map itself is constructed by taking, for each closed irreducible subvariety $Z \subset X$ of codimension $k$, the fundamental class $[Z]_{\mathrm{hom}} \in H_{2n-2k}(X, \mathbb{Z})$, applying Poincaré duality to obtain $[Z] = \mathrm{PD}([Z]_{\mathrm{hom}})$, and extending $\mathbb{Q}$-linearly from irreducible subvarieties to $Z^k(X)_{\mathbb{Q}}$.

## The integral conjecture is a different sentence

The integral Hodge conjecture asserts that

$$\mathrm{cl}_{\mathbb{Z}} \colon CH^k(X) \longrightarrow H^{2k}(X, \mathbb{Z})$$

is surjective onto $H^{2k}(X, \mathbb{Z}) \cap H^{k,k}(X)$. A class counts only if the coefficients are integers.

This sentence is not the Clay problem, and it is not open in general. Atiyah–Hirzebruch produced torsion counterexamples. Kollár produced non-torsion counterexamples on very general hypersurfaces in $\mathbb{P}^4$: integral Hodge classes which become algebraic only after multiplication by an integer greater than 1. If $m\gamma$ is algebraic for some $m \neq 0$, then $\gamma$ is algebraic over $\mathbb{Q}$. Those examples do not disprove the rational conjecture.

On the three-chain sextic $X = V(F) \subset \mathbb{P}^5$, with plane $\Pi = V(x_3, x_4, x_5)$ and residual surface $S$ satisfying $[S] = h^2 - [\Pi]$, the rational primitive projection

$$\alpha = [\Pi] - \tfrac{1}{6} h^2$$

lies in $H^4(X, \mathbb{Q})$, not a priori in $H^4(X, \mathbb{Z})$. The factor $\frac{1}{6}$ is legal for Clay and illegal as an integral class. Clearing it gives

$$\beta := 6\alpha = h^2 - 6[\Pi] = [S] - 5[\Pi].$$

Then $\langle \beta, h^2 \rangle = 0$ and $\beta$ is an integral combination of two surfaces, so $\beta$ lies in the image of $\mathrm{cl}_{\mathbb{Z}}$. Neither $\alpha$ nor $\beta$ is a miss for either conjecture.

The lattice $\mathbb{Z}h^2 + \mathbb{Z}[\Pi]$ has intersection matrix $\left(\begin{smallmatrix} 6 & 1 \\ 1 & 21 \end{smallmatrix}\right)$ and discriminant 125. The self-intersection 21 is $c_2$ of the normal bundle of the plane. Discriminant 125 leaves open whether the lattice is saturated in $H^4(X, \mathbb{Z})$. An unsaturated class would be an integral question only: it would still lie in $\mathbb{Q}\langle h^2, [\Pi] \rangle$, hence would be algebraic over $\mathbb{Q}$. No such class is produced in this note.

## Monodromy, and the number that is not a matrix group

The same lattice determines the pairings of $\beta$ and of $[S]$. With $h^4 = 6$ and $h^2 \cdot [\Pi] = 1$,

$$\begin{aligned} h^2 \cdot [S] &= 5, & [\Pi] \cdot [S] &= -20, & [S] \cdot [S] &= 25, \\ \beta \cdot h^2 &= 0, & \beta \cdot [\Pi] &= -125, & \beta \cdot [S] &= 125, \\ \beta \cdot \beta &= 750. \end{aligned}$$

These are computed from the Gram matrix. They are not a presentation of a monodromy group.

For a smooth sextic, $b_4 = 2606$, so the primitive lattice $(h^2)^\perp$ has rank 2605. A loop around a node of the universal family acts by a Picard–Lefschetz reflection on that lattice. Beauville's theorem says the group generated by those reflections is Zariski-dense in the orthogonal group of the vanishing cohomology. Deligne's invariant-cycle theorem says the invariant subspace is $\mathbb{Q}h^2$. Neither statement is a list of $2605 \times 2605$ matrices, and this note does not produce one.

Restricting the base to sextics that contain a plane replaces that group by a smaller image $\mathrm{Mon}_\Pi$. The computation above shows that $\mathrm{Mon}_\Pi$ has an invariant subspace of rank at least 2, namely $\mathbb{Q}h^2 + \mathbb{Q}[\Pi]$, and that the primitive algebraic direction $\beta$ has square 750. The index of $\mathrm{Mon}_\Pi$ is not computed. The Milnor monodromy of the plane-curve germ $u^5v + v^6$, an operator on a rank-25 lattice, is not this group.

## A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  https://github.com/BenFrohman/HODGE
- Verified commit:
  `fc6ad0982a118a824c554001ee4ad10bf401766c`
- Lean toolchain: `leanprover/lean4:v4.22.0`
- Mathlib: tag `v4.22.0`, commit (from `lake-manifest.json`)
  `79e94a093aff4a60fb1b1f92d9681e407124c2ca`
- Continuous integration runs on the verified commit (both passed): certificate workflow (build, axiom audit, `sorry`/`axiom`/`native_decide` scan)
  https://github.com/BenFrohman/HODGE/actions/runs/41731467848
  and Lean build workflow
  https://github.com/BenFrohman/HODGE/actions/runs/41731468054
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  https://github.com/BenFrohman/HODGE/archive/fc6ad0982a118a824c554001ee4ad10bf401766c.tar.gz
  `b30e388de1f1dc9d06b852622d22889813cf68bea06e5be0be9c51fea8b2d620`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

**Scope.** The audited build consists of the default `lake` targets `Hodge` (307 theorems), `HodgeAttempt` (41 theorems; a sandbox that is *not* the Clay Hodge statement and proves no Clay result) and `AxiomPrint` (axiom reports only, no theorems), 348 theorem declarations in total (238 written in the source and 110 generated automatically by Lean, such as constructor injectivity, `sizeOf` and equation lemmas). The `Hodge` root imports every module of the library except `Hodge.JacobianGenerators`, which is excluded because it redeclares, with the same names and statements, the declarations of `Hodge.SpecialSexticJacobian` (already certified) and so cannot be imported alongside it. Also excluded: `Hodge.Attempt.FermatRecipeSco` which does not compile as committed (missing $\mathbb{N}$ notation import; `decide` is stuck on $2 \le$ `offLocusGadget.codim`) and whose theorem `recipe_is_not_global_constructor` duplicates a certified declaration of the same name in `Hodge.Attempt.ParameterLocus`.

**What the Lean development does and does not prove.** The body of this note is a mathematical exposition; its theorems are cited from the literature, not proved in Lean. The table records, for each claim of the body that has a Lean counterpart, exactly what the Lean declaration states.

| **Claim in the body** | **Lean declaration** | **What Lean states** |
|---|---|---|
| Gram matrix $h^4 = 6$, $h^2 \cdot [\Pi] = 1$, $[\Pi]^2 = 21$ | `Hodge.SpecialSextic.pairing_h2_h2`, `pairing_h2_Pi`, `pairing_Pi_Pi` | Values of the bilinear form $6aa' + ab' + a'b + 21bb'$ on $\mathbb{Q}^2$, which is *defined* by this matrix. Not derived from the geometry of $X$. |
| Discriminant 125 | `Hodge.SpecialSextic.discriminant`, `Hodge.ContainerSeparation.two_plane_shadow_det` | $6 \cdot 21 - 1^2 = 125$ in $\mathbb{Q}$ (resp. $6 \cdot 21 - 1 = 125$ in $\mathbb{N}$). Saturation is not addressed. |
| $h^2 \cdot [S] = 5$, $[\Pi] \cdot [S] = -20$, $[S]^2 = 25$ | `pairing_h2_S`, `pairing_Pi_S`, `pairing_S_S` | Same form, with $S := (1, -1)$, i.e. $[S] = h^2 - [\Pi]$ taken as a definition (`S_eq`). |
| $\beta \cdot h^2 = 0$, $\beta \cdot [\Pi] = -125$, $\beta \cdot [S] = 125$, $\beta^2 = 750$ | `pairing_beta_h2`, `pairing_beta_Pi`, `pairing_beta_S`, `pairing_beta_beta` | Same form, with $\beta := (1, -6) = h^2 - 6[\Pi]$ (`beta_eq`). |
| $b_4 = 2606$ for a smooth sextic | `Hodge.HostSplit.sextic_b4`, `Hodge.ContainerSeparation.sextic_b4` | The natural-number identity $1 + 426 + 1752 + 426 + 1 = 2606$; the Hodge numbers are inputs, not computed. Rank 2605 of $(h^2)^\perp$ has no Lean statement. |
| $[\Pi]^2 = 21 = c_2(N_{\Pi/X})$ | `Hodge.Attempt.IntersectionInvariants.planeSelfIx_six` | The integer polynomial $d^2 - 3d + 3$ equals 21 at $d = 6$; no normal bundle is formalized. |
| Lefschetz $(1,1)$ settles $k = 1$ | `Hodge.hodgeConjecture_codim_one` | Conditional: *if* a datum carries the hypothesis class `LefschetzOneOne`, the formal Hodge statement holds for it. Lefschetz $(1,1)$ itself is not proved. |

| Claim in the body | Lean declaration | What Lean states |
|---|---|---|
| Hard Lefschetz settles $k = \dim X - 1$; $\dim X \leq 3$ | `Hodge.hodgeConjecture_hardLefschetz`, `Hodge.hodgeConjecture_dim_le_three` | Conditional on hypothesis classes; hard Lefschetz is not proved. |
| Rational vs. integral conjecture are different | `Hodge.Integral.rational_does_not_imply_integral`, `rational_solvable`, `integral_unsolvable` | A toy separation ($2r = q$ solvable in $\mathbb{Q}$, $2n = 1$ not in $\mathbb{Z}$); Atiyah–Hirzebruch and Kollár are not formalized. |
| The Hodge conjecture for a general fourfold is open | `Hodge.Clay.RationalHodgeCodimTwo`, `Hodge.HodgeConjecture.general_fourfold_iff_clay` | `RationalHodgeCodimTwo` is a `Prop` (a definition) with no proof in the library. The `iff` states, for an abstract linear datum of codimension 2, that `general_fourfold` is equivalent to surjectivity of cl onto the Hodge classes; neither side is proved. |

No declaration in the library proves the Hodge conjecture, Lefschetz $(1,1)$, hard Lefschetz, Beauville's or Deligne's theorem, or any monodromy statement. In the built version, `Hodge.HodgeConjecture.ungated_codim_two_false` refutes the ungated statement only for $Z, V, N$ : `Type` (universe 0); `Hodge.Release.ContainsTwoPlanes` is `Hodge.SpecialSextic.ClosedMembership.{0}`, pinned to universe 0; and `realClose_projectiveFourSpace`, `realClose_kleinQuadric`, `realClose_productOfPlanes` are definitions (data of type `RealClose`), so they carry no `#print axioms` entry. None of the claims of the body depends on these.

**The development contains no sorry and no user-declared axioms.** Every one of the 348 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

| Lean declaration | #print axioms |
|---|---|
| `Hodge.Attempt.Hypersurface.mk.inj` | `(no axioms)` |
| `Hodge.Attempt.Hypersurface.mk.injEq` | `[propext]` |
| `Hodge.Attempt.Hypersurface.mk.sizeOf_spec` | `(no axioms)` |
| `Hodge.Attempt.IntersectionInvariants.extra_class_mismatch` | `[propext, Classical.choice, Quot.sound]` |
| `Hodge.Attempt.IntersectionInvariants.isolation_not_ambient` | `[propext]` |
| `Hodge.Attempt.IntersectionInvariants.planeSelfIx_six` | `[propext]` |
| `Hodge.Attempt.IntersectionInvariants.predicted_ambient_self` | `[propext, Classical.choice, Quot.sound]` |
| `Hodge.Attempt.NamedSextic.namedSextic_eq_plane_combination` | `[propext]` |
| `Hodge.Attempt.NamedSextic.namedSextic_mem_plane` | `[propext, Quot.sound]` |
| `Hodge.Attempt.ParameterLocus.fermatLocus.inj` | `(no axioms)` |
| `Hodge.Attempt.ParameterLocus.fermatLocus.injEq` | `[propext]` |
| `Hodge.Attempt.ParameterLocus.fermatLocus.sizeOf_spec` | `(no axioms)` |
| `Hodge.Attempt.ParameterLocus.genericObstructedLocus.sizeOf_spec` | `(no axioms)` |

| Lean declaration | #print axioms |
|---|---|
| Hodge.Attempt.ParameterLocus.kleinQuadric.sizeOf_spec | (no axioms) |
| Hodge.Attempt.ParameterLocus.productOfPlanes.sizeOf_spec | (no axioms) |
| Hodge.Attempt.ParameterLocus.projectiveFourSpace.sizeOf_spec | (no axioms) |
| Hodge.Attempt.SpecialSexticMembership.F_factors | [propext] |
| Hodge.Attempt.SpecialSexticMembership.F_mem_plane | [propext, Quot.sound] |
| Hodge.Attempt.Tagged.mk.inj | (no axioms) |
| Hodge.Attempt.Tagged.mk.injEq | [propext] |
| Hodge.Attempt.Tagged.mk.sizeOf_spec | (no axioms) |
| Hodge.Attempt.Var.ofNat_toCtorIdx | (no axioms) |
| Hodge.Attempt.Var.z0.sizeOf_spec | (no axioms) |
| Hodge.Attempt.Var.z1.sizeOf_spec | (no axioms) |
| Hodge.Attempt.Var.z2.sizeOf_spec | (no axioms) |
| Hodge.Attempt.Var.z3.sizeOf_spec | (no axioms) |
| Hodge.Attempt.Var.z4.sizeOf_spec | (no axioms) |
| Hodge.Attempt.Var.z5.sizeOf_spec | (no axioms) |
| Hodge.Attempt.carriesFermatRecipe.eq_1 | (no axioms) |
| Hodge.Attempt.fermat_pair_vanishing | [propext, Quot.sound] |
| Hodge.Attempt.fermat_pair_with_remainder | [propext, Quot.sound] |
| Hodge.Attempt.fermat_quartic_contains_Z1 | [propext, Quot.sound] |
| Hodge.Attempt.fourfold_complex_dim_eq_four | [propext] |
| Hodge.Attempt.genericObstructedFourfold_codim | [propext, Classical.choice, Quot.sound] |
| Hodge.Attempt.genericObstructedTag.eq_1 | (no axioms) |
| Hodge.Attempt.generic_tag_has_no_fermat_recipe | (no axioms) |
| Hodge.Attempt.middle_cycle_codimension_is_two | [propext] |
| Hodge.Attempt.middle_cycle_dim_eq_two | [propext] |
| Hodge.Attempt.pair_vanishing | [propext, Quot.sound] |
| Hodge.Attempt.pair_with_remainder | [propext, Quot.sound] |
| Hodge.Attempt.recipe_is_not_global_constructor | (no axioms) |
| Hodge.ChowData.mk.inj | [propext, Classical.choice, Quot.sound] |
| Hodge.ChowData.mk.injEq | [propext, Classical.choice, Quot.sound] |
| Hodge.ChowData.mk.sizeOf_spec | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.P4Coord.ofNat_toCtorIdx | (no axioms) |
| Hodge.Classical.P4Coord.x0.sizeOf_spec | (no axioms) |
| Hodge.Classical.P4Coord.x1.sizeOf_spec | (no axioms) |
| Hodge.Classical.P4Coord.x2.sizeOf_spec | (no axioms) |
| Hodge.Classical.P4Coord.x3.sizeOf_spec | (no axioms) |
| Hodge.Classical.P4Coord.x4.sizeOf_spec | (no axioms) |
| Hodge.Classical.P4Cycle.mk.inj | [propext] |
| Hodge.Classical.P4Cycle.mk.injEq | [propext] |
| Hodge.Classical.P4Cycle.mk.sizeOf_spec | [propext] |
| Hodge.Classical.ProductCycle.mk.inj | [propext] |
| Hodge.Classical.ProductCycle.mk.injEq | [propext] |
| Hodge.Classical.ProductCycle.mk.sizeOf_spec | [propext] |
| Hodge.Classical.classical_islands | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.constructP4_coeff | [propext] |

| Lean declaration | #print axioms |
|---|---|
| Hodge.Classical.constructP4_discharged | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.constructP4_section | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.constructProduct_section | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.construct_recovers | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.constructions_are_sections | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.expansion | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.instCycleConstructorProdRatKleinQuadric | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.instCycleConstructorProdRatProductOfPlanes | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.instCycleConstructorRatProjectiveFourSpace | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.kleinQuadric_hodgeConjecture | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.productOfPlanes_hodgeConjecture | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.projectiveFourSpace_ hodgeConjecture | [propext, Classical.choice, Quot.sound] |
| Hodge.Classical.sections_recover | [propext, Classical.choice, Quot.sound] |
| Hodge.Clay.instNonemptyHdg2 | (no axioms) |
| Hodge.Clay.of_section_is_not_uniform | [propext, Classical.choice, Quot.sound] |
| Hodge.Clay.skeleton_on_iff | [propext, Classical.choice, Quot.sound] |
| Hodge.ContainerSeparation.Z_not_container | (no axioms) |
| Hodge.ContainerSeparation.Z_uncomputed | (no axioms) |
| Hodge.ContainerSeparation.gram_shadow_numeral | (no axioms) |
| Hodge.ContainerSeparation.quintic_h31 | (no axioms) |
| Hodge.ContainerSeparation.sextic_R12 | (no axioms) |
| Hodge.ContainerSeparation.sextic_R18 | (no axioms) |
| Hodge.ContainerSeparation.sextic_R6 | (no axioms) |
| Hodge.ContainerSeparation.sextic_b4 | (no axioms) |
| Hodge.ContainerSeparation.sextic_h22 | (no axioms) |
| Hodge.ContainerSeparation.two_plane_shadow_det | (no axioms) |
| Hodge.Cubic.ZuckerCubic.construct | [propext, Classical.choice, Quot.sound] |
| Hodge.Cubic.construct_h2_section | [propext, Classical.choice, Quot.sound] |
| Hodge.Cubic.hassettC8_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Cubic.hodge_of_zucker | [propext, Classical.choice, Quot.sound] |
| Hodge.Cubic.instCycleConstructorOfZuckerCubic | [propext, Classical.choice, Quot.sound] |
| Hodge.Cubic.no_universal_cubic_section | (no axioms) |
| Hodge.Cubic.veryGeneral_hodge | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
|---|---|
| Hodge.CycleConstructor.construct | [propext, Classical.choice, Quot.sound] |
| Hodge.CycleSection.is_section | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.algebraicClasses_le_hodgeClasses | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.cl_isHodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.hodgeClasses.eq_1 | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.hodgeConjecture_codim_blind | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.hodgeConjecture_of_generators | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.hodgeConjecture_of_integral_generation | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.mk.inj | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.mk.injEq | [propext, Classical.choice, Quot.sound] |
| Hodge.Datum.mk.sizeOf_spec | [propext, Classical.choice, Quot.sound] |
| Hodge.DerivedSurfaces.NamedDerivedSurface.mk.inj | (no axioms) |
| Hodge.DerivedSurfaces.NamedDerivedSurface.mk.injEq | [propext] |
| Hodge.DerivedSurfaces.NamedDerivedSurface.mk.sizeOf_spec | (no axioms) |
| Hodge.DerivedSurfaces.every_listed_host_is_a_fourfold | (no axioms) |
| Hodge.DerivedSurfaces.every_listed_object_is_a_surface | (no axioms) |
| Hodge.DerivedSurfaces.inventory_card | (no axioms) |
| Hodge.DerivedSurfaces.residual_host_is_fourfold | (no axioms) |
| Hodge.DerivedSurfaces.residual_is_surface | (no axioms) |
| Hodge.DerivedSurfaces.residual_not_partner | (no axioms) |
| Hodge.DimLeThree.construct | [propext, Classical.choice, Quot.sound] |
| Hodge.Examples.instCycleConstructorRatTrivialObstruction | [propext, Classical.choice, Quot.sound] |
| Hodge.Examples.not_every_codim_ge_two | [propext, Classical.choice, Quot.sound] |
| Hodge.Examples.trivialObstruction_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Examples.zeroCycle.eq_1 | [propext, Classical.choice, Quot.sound] |
| Hodge.Examples.zeroCycle_is_not_a_variety_sticker | (no axioms) |
| Hodge.Examples.zeroCycle_not_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Fermat.Coord.ofNat_toCtorIdx | (no axioms) |
| Hodge.Fermat.Coord.z0.sizeOf_spec | (no axioms) |
| Hodge.Fermat.Coord.z1.sizeOf_spec | (no axioms) |
| Hodge.Fermat.Coord.z2.sizeOf_spec | (no axioms) |
| Hodge.Fermat.Coord.z3.sizeOf_spec | (no axioms) |
| Hodge.Fermat.Coord.z4.sizeOf_spec | (no axioms) |
| Hodge.Fermat.Coord.z5.sizeOf_spec | (no axioms) |

| Lean declaration | #print axioms |
|---|---|
| Hodge.Fermat.LinearPlane.mk.inj | (no axioms) |
| Hodge.Fermat.LinearPlane.mk.injEq | [propext] |
| Hodge.Fermat.LinearPlane.mk.sizeOf_spec | (no axioms) |
| Hodge.Fermat.Pairing.mk.inj | (no axioms) |
| Hodge.Fermat.Pairing.mk.injEq | [propext] |
| Hodge.Fermat.Pairing.mk.sizeOf_spec | (no axioms) |
| Hodge.Fermat.Z1_pairing | [propext] |
| Hodge.Fermat.Z2_pairing | [propext] |
| Hodge.Fermat.construct_section | [propext, Classical.choice, Quot.sound] |
| Hodge.Fermat.contains_two_planes | [propext, Classical.choice, Quot.sound] |
| Hodge.Fermat.fermatQuartic_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Fermat.twoPlanes_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.FermatQuintic.F5_eq_G | (no axioms) |
| Hodge.FermatQuintic.G_not_euler_degree5 | [propext, Quot.sound] |
| Hodge.FermatQuintic.X_5_name_ne_T_F | (no axioms) |
| Hodge.FermatQuintic.affine_cone_isolated_at_origin | [propext, Classical.choice, Quot.sound] |
| Hodge.FermatQuintic.b4_numeral | (no axioms) |
| Hodge.FermatQuintic.canonical_degree | (no axioms) |
| Hodge.FermatQuintic.euler_homogeneous | [propext] |
| Hodge.FermatQuintic.h22_full_numeral | (no axioms) |
| Hodge.FermatQuintic.h31_numeral | (no axioms) |
| Hodge.FermatQuintic.h40_numeral | (no axioms) |
| Hodge.FermatQuintic.jacobian_certificate | [propext, Classical.choice, Quot.sound] |
| Hodge.FermatQuintic.this_is_not_general_fourfold | (no axioms) |
| Hodge.Geometry.Variety.hodgeConjecture_iff | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.Variety.mk.inj | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.Variety.mk.injEq | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.Variety.mk.sizeOf_spec | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.cycleClass_is_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.every_datum_wraps | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.kleinQuadric_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.onVarieties_false_on_Rat | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.productOfPlanes_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.projectiveFourSpace_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.wrap_data | [propext, Classical.choice, Quot.sound] |
| Hodge.Geometry.wrapped | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
|---|---|
| Hodge.Geometry.zeroGadget_not_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.HardLefschetzComplement.construct | [propext, Classical.choice, Quot.sound] |
| Hodge.Hassett.P5Coord.ofNat_toCtorIdx | (no axioms) |
| Hodge.Hassett.P5Coord.x0.sizeOf_spec | (no axioms) |
| Hodge.Hassett.P5Coord.x1.sizeOf_spec | (no axioms) |
| Hodge.Hassett.P5Coord.x2.sizeOf_spec | (no axioms) |
| Hodge.Hassett.P5Coord.x3.sizeOf_spec | (no axioms) |
| Hodge.Hassett.P5Coord.x4.sizeOf_spec | (no axioms) |
| Hodge.Hassett.P5Coord.x5.sizeOf_spec | (no axioms) |
| Hodge.Hassett.construct_section | [propext, Classical.choice, Quot.sound] |
| Hodge.Hassett.planeSpan_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.LefschetzTwoTwo_iff | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.RealClose.mk.inj | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.RealClose.mk.injEq | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.RealClose.mk.sizeOf_spec | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.classical_fourfolds | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.frohman_two_two_spans | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.frohman_two_two_spans_iff | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.general_fourfold_iff | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.general_fourfold_iff_clay | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.iff_nonempty_fibers | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.klein | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.living_skeleton | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.mem_algebraicClasses_iff | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.named_fourfolds | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.nonempty_fiber_iff | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.of_constructor | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.of_realClose | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.of_section | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.p4 | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
|---|---|
| Hodge.HodgeConjecture.product | [propext, Classical.choice, Quot.sound] |
| Hodge.HodgeConjecture.ungated_codim_two_false | [propext, Classical.choice, Quot.sound] |
| Hodge.HostSplit.Z_uncomputed | (no axioms) |
| Hodge.HostSplit.containers_differ | (no axioms) |
| Hodge.HostSplit.quintic_b4 | (no axioms) |
| Hodge.HostSplit.quintic_h22 | (no axioms) |
| Hodge.HostSplit.quintic_h31 | (no axioms) |
| Hodge.HostSplit.sextic_b4 | (no axioms) |
| Hodge.HostSplit.sextic_h22 | (no axioms) |
| Hodge.HostSplit.sextic_length | (no axioms) |
| Hodge.HostSplit.sextic_r12 | (no axioms) |
| Hodge.HostSplit.sextic_r18 | (no axioms) |
| Hodge.HostSplit.sextic_r6 | (no axioms) |
| Hodge.Integral.integral_unsolvable | [propext] |
| Hodge.Integral.rational_does_not_imply_integral | [propext, Classical.choice, Quot.sound] |
| Hodge.Integral.rational_solvable | [propext, Classical.choice, Quot.sound] |
| Hodge.Islands.P4Coord.ofNat_toCtorIdx | (no axioms) |
| Hodge.Islands.P4Coord.x0.sizeOf_spec | (no axioms) |
| Hodge.Islands.P4Coord.x1.sizeOf_spec | (no axioms) |
| Hodge.Islands.P4Coord.x2.sizeOf_spec | (no axioms) |
| Hodge.Islands.P4Coord.x3.sizeOf_spec | (no axioms) |
| Hodge.Islands.P4Coord.x4.sizeOf_spec | (no axioms) |
| Hodge.Islands.P4Cycle.mk.inj | [propext] |
| Hodge.Islands.P4Cycle.mk.injEq | [propext] |
| Hodge.Islands.P4Cycle.mk.sizeOf_spec | [propext] |
| Hodge.Islands.ProductCycle.mk.inj | [propext] |
| Hodge.Islands.ProductCycle.mk.injEq | [propext] |
| Hodge.Islands.ProductCycle.mk.sizeOf_spec | [propext] |
| Hodge.Islands.ProductSurface.P2TimesPt.sizeOf_spec | (no axioms) |
| Hodge.Islands.ProductSurface.lineTimesLine.sizeOf_spec | (no axioms) |
| Hodge.Islands.ProductSurface.ofNat_toCtorIdx | (no axioms) |
| Hodge.Islands.ProductSurface.ptTimesP2.sizeOf_spec | (no axioms) |
| Hodge.Islands.constructP4_coeff | [propext] |
| Hodge.Islands.constructP4_recovers | [propext, Classical.choice, Quot.sound] |
| Hodge.Islands.constructProduct_coeff | [propext] |
| Hodge.Islands.constructProduct_recovers | [propext, Classical.choice, Quot.sound] |
| Hodge.Islands.constructQ_coeff | [propext] |
| Hodge.Islands.constructQ_eq_initialize | [propext] |
| Hodge.Islands.constructQ_recovers | [propext, Classical.choice, Quot.sound] |
| Hodge.Islands.islands_construct | [propext, Classical.choice, Quot.sound] |
| Hodge.JacobianLength.HomogeneousCILength.dim_eq_pow | [propext] |
| Hodge.JacobianLength.five_pow_six | [propext] |
| Hodge.JacobianLength.length_num | [propext] |
| Hodge.JacobianLength.length_num_of_hyps | [propext] |

| Lean declaration | #print axioms |
|---|---|
| Hodge.JacobianLength.this_is_not_general_fourfold | (no axioms) |
| Hodge.Klein.Coord.ofNat_toCtorIdx | (no axioms) |
| Hodge.Klein.Coord.p01.sizeOf_spec | (no axioms) |
| Hodge.Klein.Coord.p02.sizeOf_spec | (no axioms) |
| Hodge.Klein.Coord.p03.sizeOf_spec | (no axioms) |
| Hodge.Klein.Coord.p12.sizeOf_spec | (no axioms) |
| Hodge.Klein.Coord.p13.sizeOf_spec | (no axioms) |
| Hodge.Klein.Coord.p23.sizeOf_spec | (no axioms) |
| Hodge.Klein.FormalCycle.mk.inj | [propext] |
| Hodge.Klein.FormalCycle.mk.injEq | [propext] |
| Hodge.Klein.FormalCycle.mk.sizeOf_spec | [propext] |
| Hodge.Klein.cycleOf_coeff | [propext] |
| Hodge.Klein.cycleOf_construct | [propext] |
| Hodge.Klein.expand | [propext, Classical.choice, Quot.sound] |
| Hodge.Klein.ideals_cover | [propext] |
| Hodge.Klein.ideals_disjoint | [propext] |
| Hodge.Klein.pairing.eq_1 | [propext, Classical.choice, Quot.sound] |
| Hodge.Klein.pairing_Pi | [propext, Classical.choice, Quot.sound] |
| Hodge.Klein.pairing_Pi' | [propext, Classical.choice, Quot.sound] |
| Hodge.Klein.pairing_cross | [propext, Classical.choice, Quot.sound] |
| Hodge.Klein.pi'_on_quadric | [propext, Classical.choice, Quot.sound] |
| Hodge.Klein.pi_on_quadric | [propext, Classical.choice, Quot.sound] |
| Hodge.Klein.pluckerEval.eq_1 | [propext, Classical.choice, Quot.sound] |
| Hodge.LefschetzOneOne.construct | [propext, Classical.choice, Quot.sound] |
| Hodge.LefschetzOneOne.is_codim_one | [propext, Classical.choice, Quot.sound] |
| Hodge.NoShadow.fermat_certificate_is_not_chow | (no axioms) |
| Hodge.NoShadow.shadow_ne_geometric_cl | (no axioms) |
| Hodge.Release.NamedTerm.mk.inj | (no axioms) |
| Hodge.Release.NamedTerm.mk.injEq | [propext] |
| Hodge.Release.NamedTerm.mk.sizeOf_spec | (no axioms) |
| Hodge.Release.Status.closed.sizeOf_spec | (no axioms) |
| Hodge.Release.Status.ofNat_toCtorIdx | (no axioms) |
| Hodge.Release.Status.open.sizeOf_spec | (no axioms) |
| Hodge.Release.contains_two_planes | [propext, Quot.sound] |
| Hodge.Release.inventory_uniform_count | (no axioms) |
| Hodge.Release.named_host_shadows | [propext, Classical.choice, Quot.sound] |
| Hodge.Release.only_uniformConstruct_is_flagged | (no axioms) |
| Hodge.Release.status_separation | (no axioms) |
| Hodge.SexticModuli.dim_PGL6 | (no axioms) |
| Hodge.SexticModuli.eq_15625_of_eq_five_pow_six | (no axioms) |
| Hodge.SexticModuli.equation_space_dim | (no axioms) |
| Hodge.SexticModuli.fixed_plane_naive_dim | (no axioms) |
| Hodge.SexticModuli.milnor_number | (no axioms) |

| Lean declaration | #print axioms |
|---|---|
| Hodge.SexticModuli.naive_git_dim | (no axioms) |
| Hodge.SexticModuli.plane_restriction_monomials | (no axioms) |
| Hodge.SexticModuli.sextic_monomials | (no axioms) |
| Hodge.SpecialSextic.F_factors | [propext] |
| Hodge.SpecialSextic.F_factors_minus1 | [propext, Quot.sound] |
| Hodge.SpecialSextic.F_mem_plane | [propext, Quot.sound] |
| Hodge.SpecialSextic.F_mem_plane_minus1 | [propext, Quot.sound] |
| Hodge.SpecialSextic.Pi.eq_1 | [propext] |
| Hodge.SpecialSextic.S.eq_1 | [propext] |
| Hodge.SpecialSextic.S_eq | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.affine_cone_isolated_at_origin | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.beta.eq_1 | [propext] |
| Hodge.SpecialSextic.beta_eq | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.closed_membership | [propext, Quot.sound] |
| Hodge.SpecialSextic.constructThree_section | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.construct_section | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.contains_two_planes | [propext, Quot.sound] |
| Hodge.SpecialSextic.dF_dx0_formula | (no axioms) |
| Hodge.SpecialSextic.dF_dx1_formula | (no axioms) |
| Hodge.SpecialSextic.dF_dx2_formula | (no axioms) |
| Hodge.SpecialSextic.dF_dx3_formula | (no axioms) |
| Hodge.SpecialSextic.dF_dx4_formula | (no axioms) |
| Hodge.SpecialSextic.dF_dx5_formula | (no axioms) |
| Hodge.SpecialSextic.discriminant | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.euler_homogeneous | [propext] |
| Hodge.SpecialSextic.five_pow_six | [propext] |
| Hodge.SpecialSextic.gradPair_origin | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.gradient_only_origin | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.gram_det_numeral | (no axioms) |
| Hodge.SpecialSextic.h2.eq_1 | [propext] |
| Hodge.SpecialSextic.head_plane_not_on_host | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.head_plane_point | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.hypersurface_contains_the_plane | [propext, Quot.sound] |
| Hodge.SpecialSextic.hypersurface_contains_the_sign_plane | [propext, Quot.sound] |
| Hodge.SpecialSextic.jacobian_lock_is_not_general_fourfold | (no axioms) |
| Hodge.SpecialSextic.jacobian_vanishes_only_at_origin | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing.eq_1 | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_Pi_Pi | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
|---|---|
| Hodge.SpecialSextic.pairing_Pi_S | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_S_S | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_beta_Pi | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_beta_S | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_beta_beta | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_beta_h2 | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_h2_Pi | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_h2_S | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.pairing_h2_h2 | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.planeSpan_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.planeSpan_identity_shadow | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.special_NL_sextic_coefficient_shadow | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.threeSpan_hodge | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSextic.threeSpan_identity_shadow | [propext, Classical.choice, Quot.sound] |
| Hodge.SpecialSexticMembership.F_factors | [propext] |
| Hodge.SpecialSexticMembership.F_factors_minus1 | [propext, Quot.sound] |
| Hodge.SpecialSexticMembership.F_mem_plane | [propext, Quot.sound] |
| Hodge.SpecialSexticMembership.F_mem_plane_minus1 | [propext, Quot.sound] |
| Hodge.SpecialSexticMembership.contains_two_planes | [propext, Quot.sound] |
| Hodge.chow_not_shadow_name | (no axioms) |
| Hodge.constructor_of_hodgeConjecture | [propext, Classical.choice, Quot.sound] |
| Hodge.hodgeConjecture_codim_one | [propext, Classical.choice, Quot.sound] |
| Hodge.hodgeConjecture_dim_le_three | [propext, Classical.choice, Quot.sound] |
| Hodge.hodgeConjecture_hardLefschetz | [propext, Classical.choice, Quot.sound] |
| Hodge.hodgeConjecture_of_constructor | [propext, Classical.choice, Quot.sound] |
| Hodge.instCycleConstructorOfCycleSection | [propext, Classical.choice, Quot.sound] |
| Hodge.instCycleConstructorOfDimLeThree | [propext, Classical.choice, Quot.sound] |
| Hodge.instCycleConstructorOfHardLefschetzComplement | [propext, Classical.choice, Quot.sound] |
| Hodge.instCycleConstructorOfLefschetzOneOne | [propext, Classical.choice, Quot.sound] |
| Hodge.instLefschetzOneOneRatK3Surface | [propext, Classical.choice, Quot.sound] |

| Lean declaration | `#print axioms` |
|---|---|
| `Hodge.k3Surface_codim` | `[propext, Classical.choice, Quot.sound]` |
| `Hodge.k3Surface_hodge` | `[propext, Classical.choice, Quot.sound]` |